

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 11 • Trigonometric Functions

Questions

Student book • questions with writing space

- 48 pages
- Past papers up to June 2026

Trigonometric Functions

Your classified route

16

questions

97

marks

4

families

11

variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Trigonometric Functions

Families and variants

F01 • Reading and identifying trigonometric graphs

V01 Key points, range and period

V02 Infer amplitude, frequency or equation from features

V03 Key points under vertical, horizontal and reflection transformations

F02 • Symmetry, reference angles and symbolic solutions

V01 Reference-angle and sign identities

V02 Solve a trigonometric equation in terms of alpha

V03 Link sine and cosine solutions by phase

F03 • Counting roots and graphical intersections

V01 Count solutions of a level equation over long intervals

V02 Count intersections with a line or non-trigonometric curve

V03 Parameter values giving an exact root count

F04 • Constructing trigonometric sketches

V01 Sketch transformed sine and cosine

V02 Sketch tangent with vertical asymptotes

Reading and identifying trigonometric graphs

WHAT THIS FAMILY ASKS

Use amplitude, period, sign, phase and key points to read a graph or infer its equation and parameters.

VARIANT ROUTE

V02 Infer amplitude, frequency or equation from features

V03 Key points under vertical, horizontal and reflection transformations

9 Complete Classified questions

54 total marks


$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

- (a) (i) State the value of n
(ii) State the period of C_1

(2)

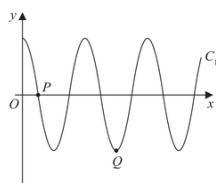


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

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The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

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The point P and the point Q lie on C_1 and are shown in Figure 3.

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

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The point P and the point Q lie on C_1 and are shown in Figure 1.

- (a) State
- (i) the coordinates of P ,
 - (ii) the coordinates of Q .

(3)

The curve C_2 has equation $y = 4\cos x^\circ + k$, where k is a constant.

Curve C_2 has a minimum y value of -1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

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$$y = A \cos(x - 30)^\circ$$

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(1)

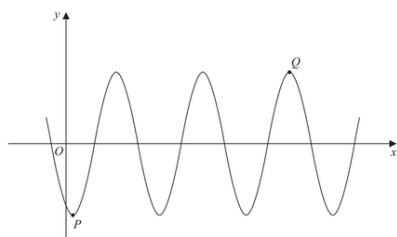


Figure 4

Figure 4 shows part of the curve with equation

$$y = A \cos(x - 30)^\circ$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q .

(3)

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9. (i)

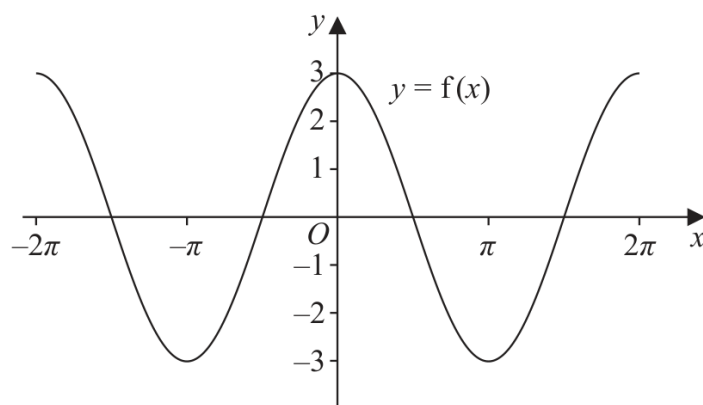


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

9. (i)

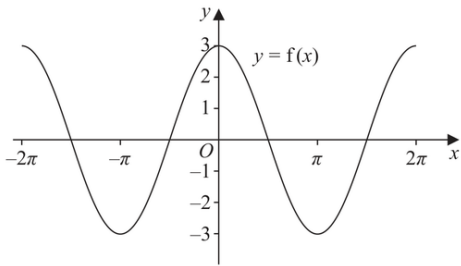


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(2)

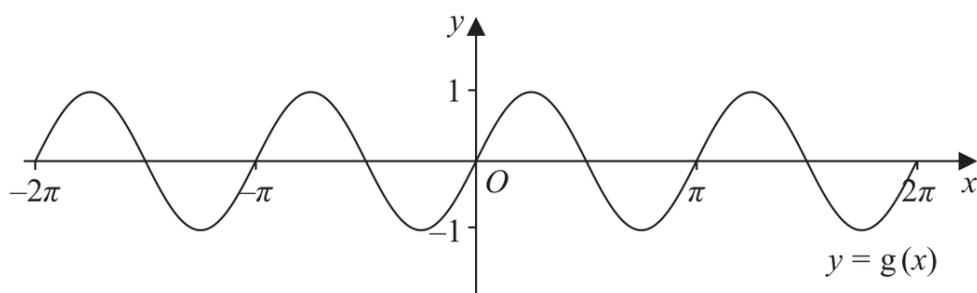


Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

- (b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

(2)

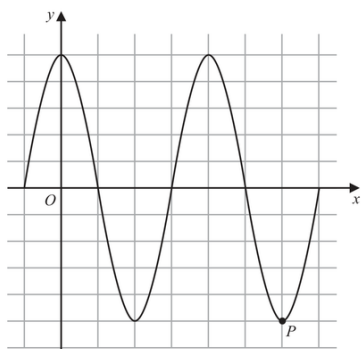


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

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Figure 4 shows a sketch of part of the curve C_1 with equation

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

- (b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

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The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

- (i) the value of A ,
- (ii) the smallest possible value of B .

(2)

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$$y = -4 \cos x$$

Points P and Q lie on the curve and are shown in Figure 3.

(ii) the coordinates of Q

24

The curve C_2 has equation $y = -4\cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

- (b) (i) state the value of k
- (ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

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On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4\cos x = 5 - \frac{10}{\pi}x$$

giving a reason for your answer.

(2)

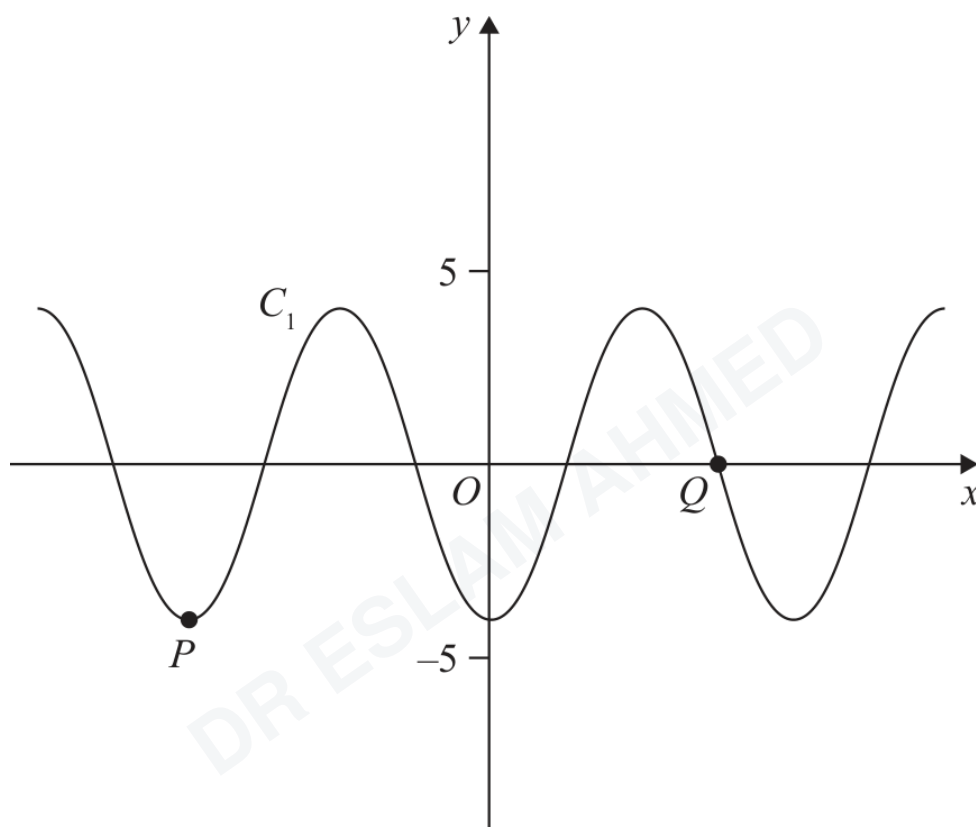


Diagram 1



Figure 4 shows a sketch of C_1 in the interval $0 \leq x < 2\pi$

(a) State the x coordinate of P .

(1)



Figure 4 shows a sketch of C_1 in the interval $0 \leq x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

$$\sin\left(x + \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

in the interval $0 \leq x < 101\pi$, giving a reason for your answer.

(2)

The curve C_2 has equation $y = -5\sin\left(x + \frac{\pi}{8}\right)$

The point Q is the minimum turning point on C_2 in the interval $0 \leq x < 2\pi$

(c) State the coordinates of Q .

(2)

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Symmetry, reference angles and symbolic solutions

WHAT THIS FAMILY ASKS

Use trigonometric symmetry and phase links to express exact values or solutions in terms of a given angle α .

VARIANT ROUTE

V02 Solve a trigonometric equation in terms of α

V03 Link sine and cosine solutions by phase

2 Complete Classified questions

12 total marks

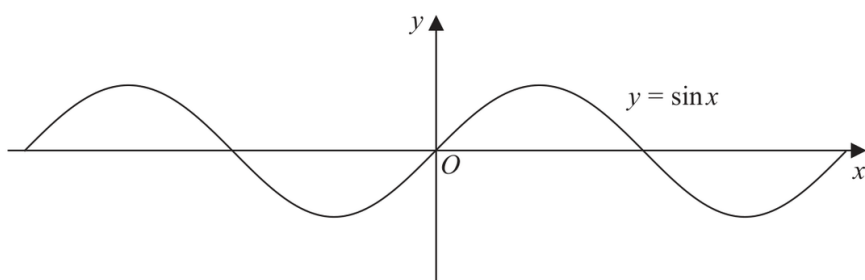


Figure 4 shows part of the graph of the curve with equation $y = \sin x$

- (a) state, in terms of p , the value of

(ii) $\sin(\alpha - 180^\circ)$

(iii) $3 + \sin(180^\circ + \alpha)$

(3)

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

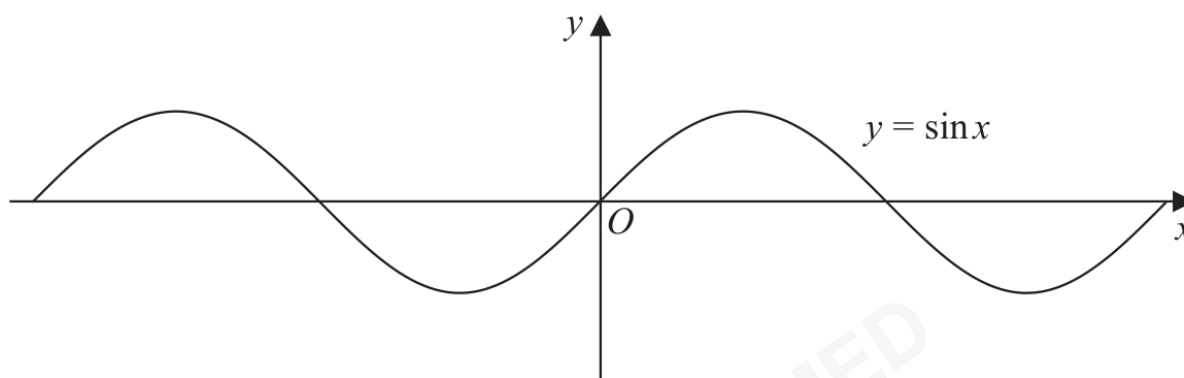


Diagram 1

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

- (c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

(3)

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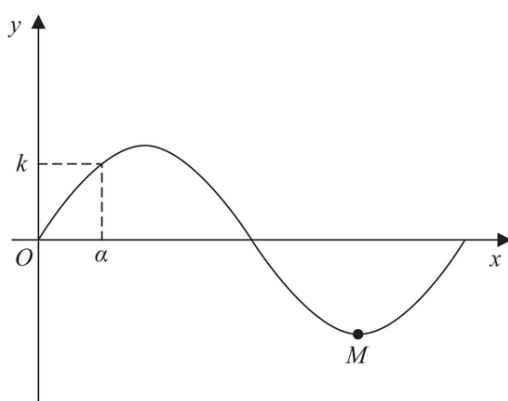


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

- (a) State the period of C .

(1)

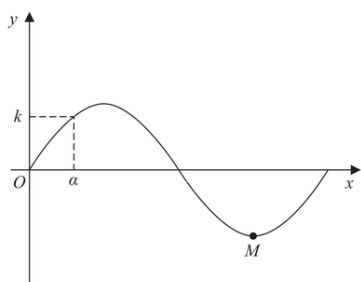


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

(b) State the coordinates of M .

(1)

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The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

- (c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,
(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$.

(2)

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Counting roots and graphical intersections

WHAT THIS FAMILY ASKS

Count solutions from periodic graphs, endpoint rules and intersections with horizontal lines or non-trigonometric curves.

VARIANT ROUTE

V01 Count solutions of a level equation over long intervals

V02 Count intersections with a line or non-trigonometric curve

V03 Parameter values giving an exact root count

5 Complete Classified questions

31 total marks


$$y = 4 \cos x$$

The points P and Q lie on the curve and are shown in Figure 1.

(1)


$$y = 4 \cos x$$

The points P and Q lie on the curve and are shown in Figure 1.

(2)

(c) State the **number** of solutions of the equation

(i) $4 \cos x = 3$ in the interval $0 < x \leq 18\,000^\circ$

(ii) $5 + 4 \cos x = 1$ in the interval $-720^\circ < x \leq 720^\circ$

(iii) $4 \cos x - 3 = 1$ in the interval $-1080^\circ \leq x \leq 1080^\circ$

(3)

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$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

(a) State an equation for l .

(1)

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

(4)

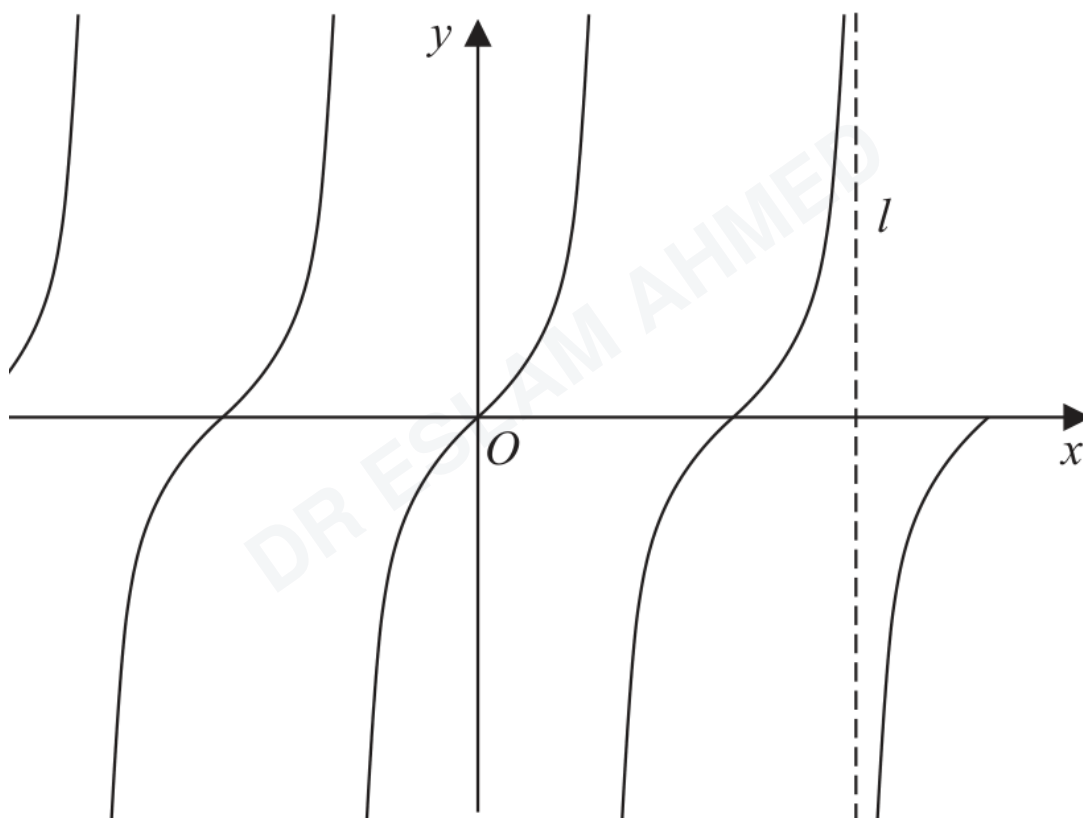


Diagram 1

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

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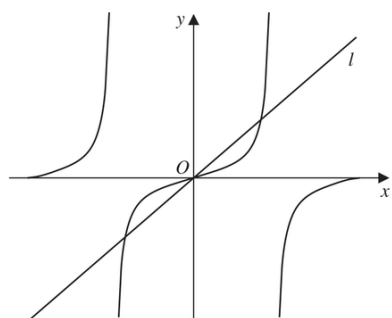


Figure 3

Figure 3 shows a sketch of

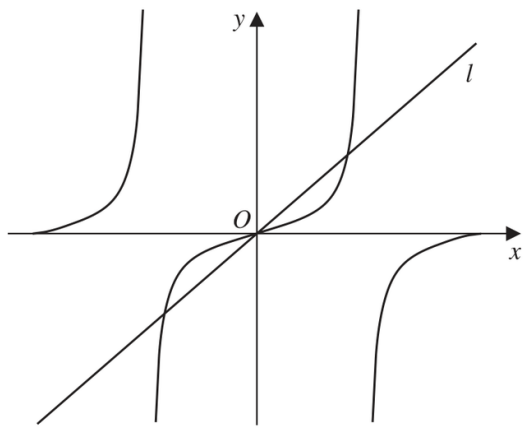
- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

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(b) Write down the number of roots of the equation

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

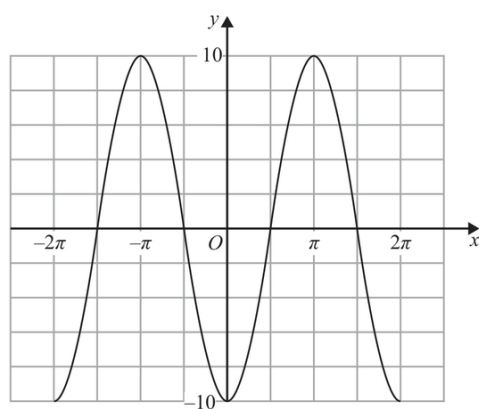


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$.

(a) Write down an expression for $f(x)$.

(2)

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Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval} \quad -100\pi \leq x \leq 100\pi$$

(3)

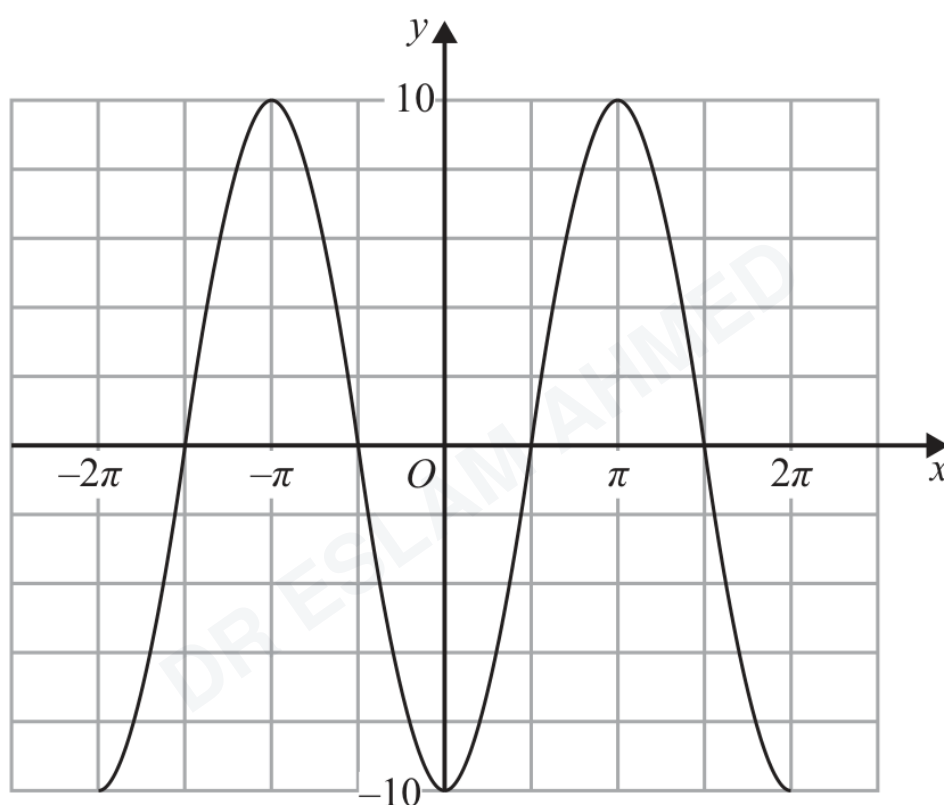


Diagram 1

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = \tan x \quad \text{in the interval} \quad -100\pi \leq x \leq 100\pi$$

giving a reason for your answer.

(4)

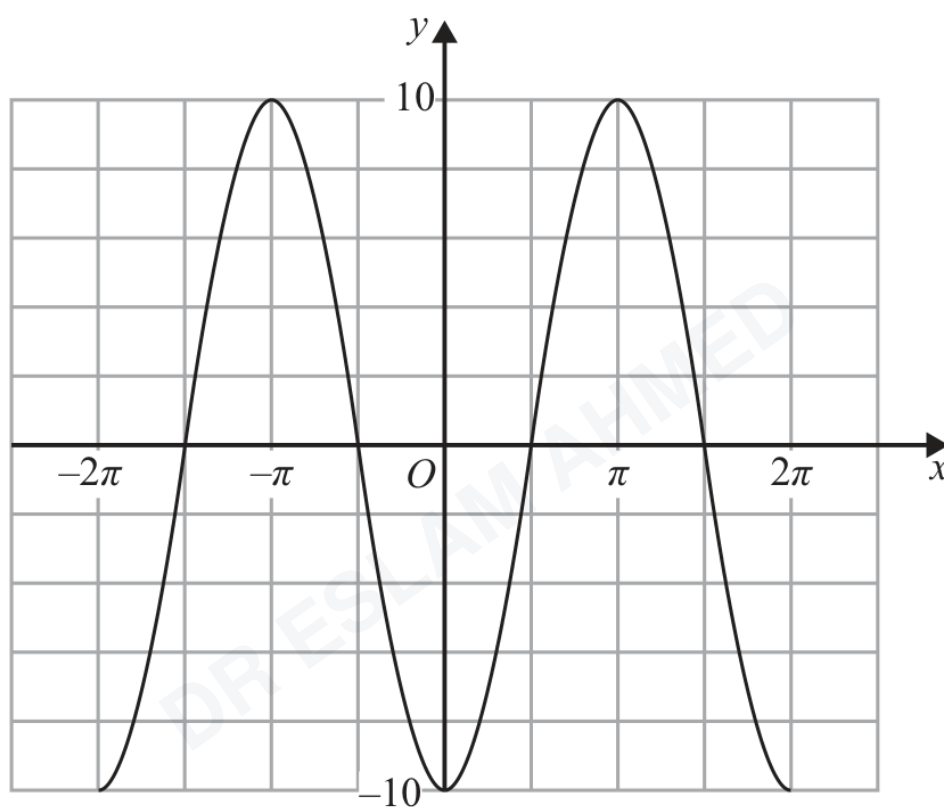


Diagram 2

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

- (i) the coordinates of Q ,
- (ii) the value of k .

(b) Given that there are exactly two solutions to the equation

$$\cos 2x^\circ = p \quad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p .

(2)

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